The Economic Impact of Broadband Infrastructure Investment

Evidence from the USDA ReConnect Program

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Motivation

- **Broadband is essential infrastructure** for 21st century economic development
- **The digital divide:** Before pandemic, 20+ million Americans lacked reliable internet
- **Major policy response:** Infrastructure Investment and Jobs Act (2021)
  - $65 billion allocated for broadband deployment
  - $42.45 billion for BEAD program (ongoing)
  - $2+ billion through USDA ReConnect (completed projects)
- **Key question:** Do these investments actually improve local economic outcomes?

Research Question

Main Question

Did counties receiving ReConnect broadband funding achieve better economic outcomes than similar counties without funding?

Focus:

- 959 counties across 10 states
- Time period: 2019 (pre) to 2022-2023 (post)
- Main outcomes: Employment and household income
- Method: Difference-in-differences

Preview of Main Finding

Surprising Result

ReConnect funding led to a **statistically significant reduction** of 764 jobs in treated counties relative to controls ($p < 0.01$)

But what does this mean?

- Not absolute job losses!
- Treated counties grew **slower** than controls
- Control counties: +4.2% employment growth
- Treated counties: +2.6% employment growth
- Suggests short-run adjustment costs

The ReConnect Program

USDA ReConnect Program (launched 2019):

- Provides grants and loans for rural broadband infrastructure
- Targets areas lacking sufficient internet access ($<$ 25 Mbps)
- Awards announced 2019-2022, deployment 2020-2023
- Over $2 billion awarded to rural communities

Policy Relevance:

- ReConnect shares design features with BEAD program
- Both prioritize unserved rural areas
- Evidence from ReConnect informs $42.45 billion BEAD deployment

Data Sources

Variable	Source
Employment	BLS Quarterly Census of Employment & Wages (QCEW) - covers 95% of all jobs
Income	Census American Community Survey (ACS) Limited to 114 counties (data availability)
Treatment	USDA ReConnect award announcements 2019-2022 funding rounds
Rural Status	USDA Rural-Urban Continuum Codes (RUCC)

Final Sample:

- 959 counties in 10 states (VA, NC, GA, TN, KY, MI, WI, MN, MT, ID)
- 122 treated, 837 control counties
- 3 time periods: 2019, 2022, 2023

Empirical Strategy: Difference-in-Differences

Basic DiD Specification:

$$Y_{it} = \alpha_i + \lambda_t + \beta \cdot (\text{Treated}_i \times \text{Post}_t) + \varepsilon_{it}$$

Where:

- Y_{it} = Employment (or income) in county i, year t
- α_i = County fixed effects (control time-invariant differences)
- λ_t = Year fixed effects (control common time trends)
- β = **Treatment effect** (our coefficient of interest)

Standard errors: Clustered at county level

Key Identification Assumption: Parallel Trends

Parallel Trends Assumption

In the absence of treatment, treated and control counties would have experienced the same employment trends

How we test this:

1. Visual inspection of pre-treatment trends (Figure 1)
2. Event study showing no effects in 2019
3. Baseline balance tests (Table 8)

Limitation: Only one pre-treatment period (2019)

- ReConnect started in 2019 - can't use earlier years
- Rely on visual evidence and baseline balance

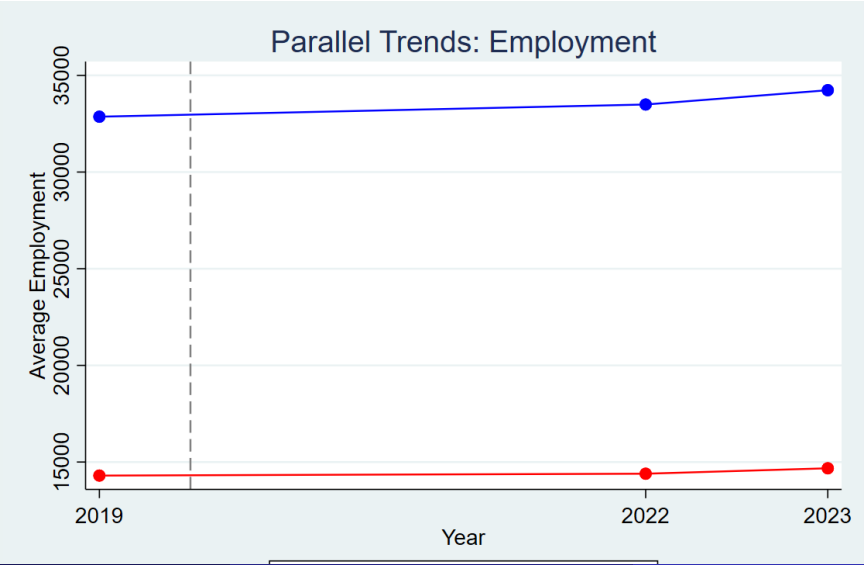
Summary Statistics (2019 Baseline)

	Control (N=837)			Treated (N=122)		
Variable	Mean	Std. Dev.	N	Mean	Std. Dev.	N
Total Employment	32,864	87,776	837	14,300	25,073	122
Median HH Income	$76,513	$16,871	294	$76,551	$8,533	29
Establishments	2,099	5,935	837	962	1,733	122
Rural (RUCC $\geq$ 4)	0.579	0.494	837	0.746	0.437	122

Key observations:

- Treated counties significantly smaller (14,300 vs. 32,864 employees)
- More rural (75% vs. 58% rural)
- **Similar income levels** ($76,551 vs. $76,513) - no systematic economic differences
- County fixed effects will control for size and rural status

Visual Evidence: Parallel Trends (Employment)



Main Result: Employment Effects

	(1) Basic	(2) w/ Rural	(3) State-Year FE	(4) Log Spec
Treatment × Post	-763.92***	-763.92***	1785.04	-0.0062
	(278.91)	(278.91)	(1838.27)	(0.0075)
Observations	2,877	2,877	318	2,877
Adjusted R^2	0.998	0.998	0.999	0.999

Interpretation of Column (1):

- **764 fewer jobs** in treated counties ($p = 0.006$)
- Represents 5.3% reduction relative to baseline
- Highly statistically significant at 1% level

What Does "Negative Effect" Mean?

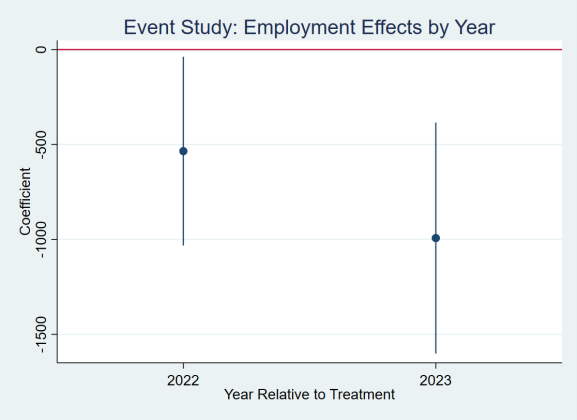
Year	Control Avg	Treated Avg	Difference
2019	32,864	14,300	18,564
2022	33,495	14,396	19,099
2023	34,234	14,678	19,556
Growth	**+1,370 (+4.2%)**	**+378 (+2.6%)**	**Gap +992**

Critical Insight

Not absolute job losses! Both groups added jobs, but treated counties grew more slowly than controls. The negative effect represents **slower employment growth**, not economic collapse.

Event Study: Effects Over Time

Year	Effect
2022	-535**
	(253)
2023	-993***
	(310)



Key findings:

- Effect appears immediately (2022)
- Effect **grows** over time
- Nearly doubles by 2023

Interpretation: Cumulative impact as more households adopt broadband and behavioral adjustments occur

Income Effects

	(1) Basic	(2) w/ Rural	(3) State-Year FE	(4) Log Spec
Treatment × Post	97.63	97.63	1.97	0.0023
	(945.43)	(945.43)	(1212.79)	(0.0138)
Observations	323	323	318	323
Counties	114	114	112	114

Interpretation:

- Small effect (+$98), not statistically significant
- Sample limited to 114 counties - reduced power
- Suggests no detectable income impact
- Consistent with job displacement (residents keep income via remote work)

Heterogeneity: Rural vs. Urban Counties

	Employment	Income
Treatment × Post × Rural	-969.60***	465.18
	(240.70)	(455.25)
Treatment × Post × Urban	-160.16	20.23
	(601.84)	(1102.32)
Test: Rural = Urban (p-value)	0.145	0.659

Key finding:

- **Negative effect concentrated in rural counties**
- Rural: -970 jobs (highly significant)
- Urban: -160 jobs (not significant)
- Suggests rural labor markets more vulnerable to displacement

Robustness Checks

	Drop Outliers	State Cluster	Common Sample	Establish-ments
Treatment × Post	-394*	-404	-404	-332***
	(226)	(2399)	(2624)	(70)
Observations	2,598	323	323	2,877

Interpretation:

- Consistently negative across specifications
- **Establishments also declined** (-332, highly significant)
- Not just employment - businesses closing too
- Results robust to outliers, clustering, sample restrictions

Why Negative Employment Effects?

Four potential mechanisms:

1. **Remote Work Displacement**
   - Broadband enables residents to work remotely for firms outside county
   - Measured employment falls even though residents remain employed
   - Especially plausible in rural areas with limited local options
2. **Short-Run Disruption**
   - Infrastructure construction temporarily disrupts businesses
   - But growing effects over time somewhat inconsistent with this
3. **Sectoral Reallocation (Creative Destruction)**
   - Broadband enables e-commerce, displacing brick-and-mortar retail
   - Traditional jobs disappear quickly, new digital jobs emerge slowly
   - Net effect negative in short run
4. **Selection and Pre-Existing Trends**
   - ReConnect may have targeted declining counties
   - But parallel trends evidence argues against this

Most likely: Combination of remote work displacement and sectoral reallocation

Policy Implications

Key takeaways for policymakers:

1. **Don't assume automatic positive effects**
   - Infrastructure investment $\neq$ guaranteed local job growth
   - Short-run adjustment costs are real
2. **Need complementary policies**
   - Workforce development programs
   - Support for businesses adapting to digital economy
3. **Rural areas require different approaches**
   - Effects concentrated in rural counties
   - Need targeted interventions for rural labor markets
4. **Long-run evaluation needed**
   - Effects may be temporary (1-4 years observed)
   - Positive effects might emerge over 5-10 years
5. **Implications for $42.45B BEAD Program**
   - Anticipate short-run adjustment costs
   - Plan complementary workforce and business support

Limitations

Key limitations to acknowledge:

1. **Limited time periods**
   - Only 2019, 2022, 2023 observed
   - Can't test pre-trends with multiple pre-periods
2. **Income data limited**
   - Only 114 of 959 counties have income data
3. **Couldn't test all proposed mechanisms**
   - Wanted deployment vs. adoption - lacked data
   - Wanted sector-specific effects - data unavailable
4. **Can't observe actual broadband deployment timing**
   - Treatment based on award dates
   - Actual deployment may lag 12-24 months
5. **Can't distinguish job losses from remote work shifts**
   - County data counts jobs by workplace location
   - Complicates welfare interpretation

Despite limitations: DiD design is sound, results are robust, findings are policy-relevant

Summary of Contributions

This paper contributes to literature in three ways:

1. **Rigorous causal estimates of ReConnect**
   - Quasi-experimental DiD methodology
   - 959 counties across 10 states, 2019-2023
   - Evidence directly relevant for $42.45B BEAD program
2. **Documents unexpected negative short-run effects**
   - 764 fewer jobs in treated counties (slower growth)
   - Highlights adjustment costs often ignored in policy debates
   - Suggests need for complementary policies
3. **Demonstrates heterogeneous effects**
   - Effects concentrated in rural counties (-970 vs. -160)
   - Broadband's impact depends on local economic conditions
   - Rural areas need different policy approaches

Main Takeaway

Key Message

Broadband infrastructure investment does **not automatically generate local employment gains**. The path from infrastructure to economic development is complex, with potential short-run adjustment costs that policymakers must anticipate and address.

Implications for $65 billion in federal broadband funding:

- Plan for transition costs
- Implement complementary workforce development
- Tailor policies to rural vs. urban contexts
- Monitor long-run outcomes (5-10 years)
- Don't expect immediate job creation

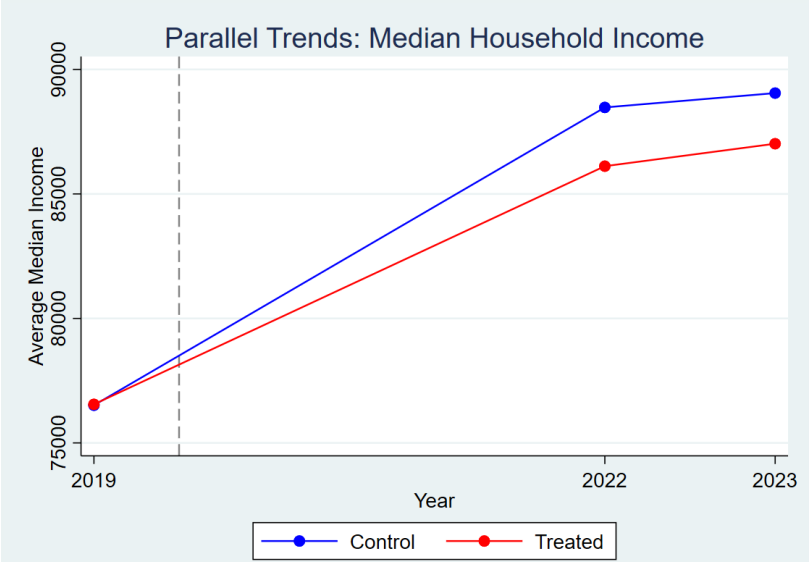
Future Research

Important directions for future work:

1. **Longer time horizons**
   - Track ReConnect counties over 5-10 years
   - Test if negative effects reverse in long run
2. **Worker-level data**
   - Distinguish job losses from remote work shifts
   - Better welfare interpretation
3. **Sector-specific analysis**
   - Which industries most affected?
   - Test retail vs. professional services vs. manufacturing
4. **Deployment vs. adoption mechanisms**
   - Does infrastructure alone drive effects?
   - Major policy implications for subsidies
5. **Complementary policies**
   - Test effectiveness of workforce development programs
   - Identify best practices for minimizing adjustment costs

Questions?

Backup: Income Parallel Trends



Backup: Full Regression Output

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Treatment × Post	-763.92***	-763.92***	1785.04	-0.0062
	(278.91)	(278.91)	(1838.27)	(0.0075)
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	Yes
State × Year FE	No	No	Yes	No
Observations	2,877	2,877	318	2,877
Adjusted R^2	0.998	0.998	0.999	0.999
Clusters (counties)	959	959	112	959

Notes: Dependent variable is total county employment (Columns 1-3) or log employment (Column 4). Standard errors clustered at county level. *** $p<0.01$, ** $p<0.05$, * $p<0.10$.